

- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! Outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*, 2023.
- Xinyang Hu, Fengzhuo Zhang, Siyu Chen, and Zhuoran Yang. Unveiling the statistical foundations of chain-of-thought prompting methods. *arXiv preprint arXiv:2408.14511*, 2024.
- Muhammed Emrullah Ildiz, Yixiao Huang, Yingcong Li, Ankit Singh Rawat, and Samet Oymak. From self-attention to Markov models: unveiling the dynamics of generative transformers. In *International Conference on Machine Learning*, 2024.
- Martin Nilsson Jacobi. A robust spectral method for finding lumpings and meta-stable states of non-reversible Markov chains. *Electronic Transactions on Numerical Analysis*, 37:296–306, 2010.
- Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, et al. OpenAI o1 system card. *arXiv preprint arXiv:2412.16720*, 2024.
- Ziwei Ji and Matus Telgarsky. Risk and parameter convergence of logistic regression. *arXiv preprint arXiv:1803.07300*, 2019.
- Andy L Jones. Scaling scaling laws with board games. *arXiv preprint arXiv:2104.03113*, 2021.
- Daniel Kahneman. Thinking, fast and slow. *Farrar, Straus and Giroux*, 2011.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- Michael Kearns. Efficient noise-tolerant learning from statistical queries. *Journal of the ACM*, 45(6):983–1006, November 1998.
- Juno Kim and Taiji Suzuki. Transformers provably solve parity efficiently with chain of thought. *arXiv preprint arXiv:2410.08633*, 2024.
- Team Kimi, Angang Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, Chenjun Xiao, Chenzhuang Du, Chonghua Liao, et al. Kimi k1.5: scaling reinforcement learning with LLMs. *arXiv preprint arXiv:2501.12599*, 2025.
- Aviral Kumar, Vincent Zhuang, Rishabh Agarwal, Yi Su, John D Co-Reyes, Avi Singh, Kate Baumli, Shariq Iqbal, Colton Bishop, Rebecca Roelofs, et al. Training language models to self-correct via reinforcement learning. *arXiv preprint arXiv:2409.12917*, 2024.

- C. Landim. Metastability for a non-reversible dynamics: The evolution of the condensate in totally asymmetric zero range processes. *Communications in Mathematical Physics*, 330, 2012.
- C. Landim. Metastable Markov chains. *arXiv preprint arXiv:1807.04144*, 2018.
- Claudio Landim and Tiecheng Xu. Metastability of finite state Markov chains: A recursive procedure to identify slow variables for model reduction. *Latin American Journal of Probability and Mathematical Statistics*, 13, 2015.
- David Asher Levin, Yuval Peres, and Elizabeth Lee Wilmer. *Markov Chains and Mixing Times*. American Mathematical Society, 2nd edition, 2009.
- Hongkang Li, Meng Wang, Songtao Lu, Xiaodong Cui, and Pin-Yu Chen. How do nonlinear transformers acquire generalization-guaranteed CoT ability? In *High-dimensional Learning Dynamics 2024: The Emergence of Structure and Reasoning*, 2024a.
- Yingcong Li, Kartik Sreenivasan, Angeliki Giannou, Dimitris Papailiopoulos, and Samet Oymak. Dissecting chain-of-thought: compositionality through in-context filtering and learning. In *Advances in Neural Information Processing Systems*, 2023.
- Zhiyuan Li, Hong Liu, Denny Zhou, and Tengyu Ma. Chain of thought empowers transformers to solve inherently serial problems. *arXiv preprint arXiv:2402.12875*, 2024b.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- Neal Madras and Dana Randall. Markov chain decomposition for convergence rate analysis. *Annals of Applied Probability*, 12, 2001.
- Ashok Vardhan Makkuva, Marco Bondaschi, Adway Girish, Alliot Nagle, Martin Jaggi, Hyeji Kim, and Michael Gastpar. Attention with Markov: A framework for principled analysis of transformers via Markov chains. *arXiv preprint arXiv:2402.04161*, 2024.
- William Merrill and Ashish Sabharwal. The expressive power of transformers with chain of thought. *arXiv preprint arXiv:2310.07923*, 2023.
- Carl D. Meyer. Stochastic complementation, uncoupling Markov chains, and the theory of nearly reducible systems. *SIAM Review*, 31(2):240–272, 1989.
- Carl Dean Meyer. The condition of a finite Markov chain and perturbation bounds for the limiting probabilities. *SIAM J. Algebraic Discret. Methods*, 1:273–283, 1980.
- Eshaan Nichani, Alex Damian, and Jason D. Lee. How transformers learn causal structure with gradient descent. *arXiv preprint arXiv:2402.14735*, 2024.
- Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, Jacob Austin, David Bieber, David Dohan, Aitor Lewkowycz, Maarten Bosma, David Luan, et al. Show your work: scratchpads for intermediate computation with language models. *arXiv preprint arXiv:2112.00114*, 2021.

- OpenAI. Spinning up: proximal policy optimization (PPO), 2018. URL <https://spinningup.openai.com/en/latest/algorithms/ppo.html>. Accessed: 2025-01-26.
- Daniel Paulin. Concentration inequalities for Markov chains by Marton couplings and spectral methods. *Electronic Journal of Probability*, 20:1–32, 2015.
- Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training. *OpenAI Blog*, 2018.
- Clayton Sanford, Bahare Fatemi, Ethan Hall, Anton Tsitsulin, Mehran Kazemi, Jonathan Halcrow, Bryan Perozzi, and Vahab Mirrokni. Understanding transformer reasoning capabilities via graph algorithms. *arXiv preprint arXiv:2405.18512*, 2024a.
- Clayton Sanford, Daniel Hsu, and Matus Telgarsky. Transformers, parallel computation, and logarithmic depth. In *International Conference on Machine Learning*, 2024b.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Shai Shalev-Shwartz, Ohad Shamir, and Shaked Shammah. Failures of gradient-based deep learning. In *International Conference on Machine Learning*, 2017.
- Ohad Shamir. Distribution-specific hardness of learning neural networks. *Journal of Machine Learning Research*, 19:32:1–32:29, 2018.
- Kumar Shridhar, Alessandro Stolfo, and Mrinmaya Sachan. Distilling reasoning capabilities into smaller language models. *arXiv preprint arXiv:2212.00193*, 2022.
- David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dhharshan Kumaran, Thore Graepel, et al. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science*, 362(6419):1140–1144, 2018.
- Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.
- Ryan Tifenbach. On an SVD-based algorithm for identifying meta-stable states of Markov chains. *Electronic Transactions on Numerical Analysis*, 38:17–33, 2011.
- Trieu H Trinh, Yuhuai Wu, Quoc V Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625(7995):476–482, 2024.
- Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. Solving math word problems with process-and outcome-based feedback. *arXiv preprint arXiv:2211.14275*, 2022.